MIL-I-12637B(EL)
22 January 1971

SUPERSEDING
MIL-I-12637A(SC)
12 NOVEMBER 1958

MILITARY SPECIFICATION

INTERCOMMUNICATION STATIONS LS-124( )/F1, LS-125( )/F1 LS-126( )/F1, LS-127( )/F1, LS-128( )/F1, LS-138( )/F1 LS-200( )/F1, LS-201( )/F1, LS-202( )/F1

1. SCOPE

1.1 This specification covers nine types of master stations for loudspeaker intercommunication using wire transmission. The intercommunication sets are designed to permit two way communication between two master stations or a master station and a slave station of compatible types. When using master stations alone in an intercommunication system, switches and controls may be manipulated to permit interconnection of each other in any combination desired. All sets except the LS-138( )/F1 utilize 115 volts a.c. as initial operating voltage. The LS-138( )/F1 utilizes either 115V or 230V for operating power. Table 1 depicts the various characteristics of any given type of intercommunication set covered by this specification. These items are designated as Intercommunication Stations LS-124( )/F1, LS-125( )/F1, LS-126( )/F1, LS-127( )/F1, LS-128( )/F1, LS-138( )/F1, LS-200( )/F1, LS-201( )/F1, LS-202( )/F1. (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification:

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
QQ-S-781	Steel Strapping, Flat
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood

FSC-5830

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SPECIFICATIONS

FEDERAL

PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock(Container Grade), and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping
PPP-T-45	Tape, Gummed, Paper, Reinforced
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing)
PPP-T-97	Tape Pressure Sensitive

MILITARY

MIL-E-75	Electron Tubes
MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material
MIL-P-3938	Pallet, Material Handling
MIL-P-11268	Parts, Materials, and Processes Used in Signal Corps Equipment
MIL-M-13189	Microphone Element, Carbon and Earphone Element, Magnetic
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-15011	Pallet, Material Handling
MIL-C-45662	Calibration of Standards

STANDARDS

FEDERAL

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Tape, Pressure Sensitive and Gummed
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment

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DRAWINGS

ELECTRONICS COMMAND

SC-DL-92315	Intercommunication Station LS-124/F1
SC-DL-108466	Intercommunication Station LS-127/F1
SC-DL-135849	Intercommunication Station LS-126/F1
SC-DL-140061	Intercommunication Station LS-125/F1
SC-DL-145852	Intercommunication Station LS-138/F1
SC-DL-430176	Intercommunication Station LS-201/F1
SC-DL-545125	Intercommunication Station LS-200/F1
SC-DL-545200	Intercommunication Station LS-202/F1

3. REQUIREMENTS

3.1 Preproduction Samples - The contractor shall furnish preproduction samples for approval as required by the invitation for bids and contract(See 6.1f). When preproduction samples are not specified by contract or purchase order, five(5) production units of the first 20 units produced shall be tested to the same requirements as specified for preproduction samples under 4.3(See 4.5.3.1).

3.2 Construction -

3.2.1 The following Intercommunication Station Sets shall be constructed in accordance with the drawings on their respective drawing and data lists including MIL-P-11268 requirements for parts, materials and processes and the requirements of this specification:

LS-124( )/F1	SC-DL-92315
LS-125( )/F1	SC-DL-140061
LS-126( )/F1	SC-DL-135849
LS-127( )/F1	SC-DL-108466
LS-138( )/F1	SC-DL-145852
LS-200( )/F1	SC-DL-545125
LS-201( )/F1	SC-DL-430176
LS-202( )/F1	SC-DL-545200

3.2.2 Intercommunication Station LS-128( )/F1 -

3.2.2.1 Procurement Model - A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

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3.2.2.2 Physical Construction - Physical construction of the equipment shall conform to the model.

3.2.2.3 Features - The equipment shall incorporate all features of the model.

3.2.2.4 Performance - The contractor shall adjust and aline the model for optimum performance and then shall measure its performance characteristics. Performance of the equipment shall conform to those measurements, except that in case of conflict between specified performance characteristics and the performance of the model, specified performance characteristics shall govern. The contractor shall immediately prepare copies of a test report on the measurements, and forward it as required by the contract.

3.2.3 Parts, Materials, and Processes; General - Except as otherwise specified herein, parts, materials, and processes shall conform to MIL-P-11268 including the selection requirements therein.

3.2.4 Finish - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings(See 4.4).

3.2.5 Marking -

3.2.5.1 General - Marking shall conform to MIL-M-13231, the equipment drawings and the model(See 4.4).

3.2.5.2 Serial Numbers - The units listed in 1.1 shall have serial numbers.

3.3 Tropicalization - Tropicalization of the equipment is required in accordance with MIL-P-11268.

3.3.1 Statement of Treatment - The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use(See 6.1i).

3.4 Performance -

3.4.1 Amplifier -

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TABLE I

INTERCOMMUNICATION STATIONS-GENERAL DESCRIPTION

Equipment	Selection Positions No. of Stations	Handset	Annun-ciator & Buzzer	Nominal Oper Voltage	Sta. Sel. Switch (No. of Positions)	Junc. Boxes (No.)	Term Strips In Box	Lugs. Pairs Per Strip	Supernant Cable No.
LS-124( )/F1	12	No	No	115	2	1	1	14	9513
LS-125( )/F1	12	No	Yes	115	3	1	2	14	9515
LS-126( )/F1	6	No	No	115	2	1	1	8	9512
LS-127( )/F1	24	No	Yes	115	3	2	2	14	9515
LS-128( )/F1	24	No	No	115	2	1	2	14	9514
LS-138( )/F1	12	No	Yes	115 or 230	3	1	2	14	9513
LS-200( )/F1	6	Yes	Yes	115	3	1	1	14	9529
LS-201( )/F1	12	Yes	Yes	115	3	1	2	14	9515
LS-202( )/F1	24	Yes	Yes	115	3	2	2	14	9515

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3.4.1.1 Listen Position(Privacy Line) - The frequency response shall be flat within -5 db to +1 db from 300 to 4000 Hz. The 1000 Hz gain shall not exceed +5 db nor fall below -2 db. With 2 watts output the total harmonic distortion shall not exceed 13 percent over the frequency range of 300 to 6000 Hz.

3.4.1.2 Listen Position - The frequency response shall fall within the limits of -6 db to +1 db from 200 to 8000 Hz. The 1000 Hz gain shall not be less than 64 db. With 2 watts output the total harmonic distortion shall not exceed 13 percent over the frequency range of 300 to 6000 Hz.

3.4.1.3 Talk Position - The frequency response shall fall within the limits of -6 db to +1 db from 200 to 8000 Hz. The 1000 Hz gain shall be not less than 64 db. With 2 watts output the total harmonic distortion shall not exceed 13 percent over the frequency range of 300 to 6000 Hz.

3.4.1.4 Total Noise - For any position of the volume control the total noise shall not exceed 0.10 volts.

3.4.1.5 Input Power - The input power shall not exceed 30 watts for any position of the listen-talk switch.

3.4.2 Loudspeaker - The loudspeaker shall have a capacity of at least 2 watts over the frequency range of 300 to 6000 Hz.

3.4.2.1 Voice Coil Impedance - The 1000 Hz voice coil impedance shall be 45 ±5 ohms.

3.4.2.2 Response - The average sound pressure response shall be not less than 95 db referred to .0002 dynes per square centimeter over the range of 300 to 6000 Hz. The response shall be flat within 14 db over this frequency range and shall have no excessive peaks or drops.

3.4.2.3 Harmonic Distortion - Total harmonic distortion shall not exceed 5 percent over the frequency range of 300 to 6000 Hz.

3.4.2.4 Microphone Output - When tested as a microphone in accordance with 4.9.2.4, the loudspeaker shall develop the voltages indicated below.

Frequency Range(Hz)	Minimum Voltage(Millivolts)
300-500	5.5
500-1000	2.5
1000-6000	0.75

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3.4.2.5 Dielectric Strength - The loudspeaker shall be capable of meeting the requirements of 3.4.2.1 thru 3.4.2.4 after being subjected to the test of 4.9.2.5.

3.4.3 Handset - The handset elements shall conform to MIL-M-13189. The receiver element shall have an impedance of 135 ohms at 1000 Hz and the transmitter element shall have an impedance of between 15 and 30 ohms at 1000 Hz.

NOTE: The contractor/vendor shall supply a certificate of compliance that the loudspeaker(3.4.2) and the handset(3.4.3) including the cradle conform to all applicable requirements of this specification including MIL-P-11268, Tropicalization and Service Conditions.

3.4.4 Buzzer - The buzzer shall conform to the tests of 4.9.3.

3.4.5 Overall Operation - The equipment shall function satisfactorily, both mechanically and electrically. This includes all voice and signal components.

3.5 Service Conditions - The equipment shall meet the following service conditions, where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.5.1 Operation - Continuous use for a period of one(1) year, under the environmental conditions specified in 3.5.2 and 3.5.3 with a duty cycle of 12 hours on and 12 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, and processes, construction methods, etc., he shall be guided by this requirement.

3.5.2 Temperature(See 4.10) -

a. Operating - Ambient temperature in the range of +150°F to -25°F(the 150°F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

b. Non-Operating - Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours, at any one time.

3.5.3 Relative Humidity(See 4.11) - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation for 4 hours.

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3.6 Preconditioning - Items shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7.

3.7 Interchangeability - Like units, assemblies, subassemblies and replacable parts shall conform to requirement 7 of MIL-STD-454. (See 4.12).

3.8 Workmanship - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - Inspection shall be classified as follows:

a. Preproduction inspection (does not include reparation for delivery) (See 4.3).

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection.

d. (1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality Conformance inspection of preparation for delivery (See 4.13).

4.3 Preproduction Inspection - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (See Tables II and III, respectively).

4.4 Inspection Covered by Subsidiary Documents - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

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Item	Where Required
Finish	3.2.4
Marking	3.2.5

4.5. Quality Conformance Inspection of Equipment Before Preparation for Delivery - Each unit of each lot of equipment shall be subjected to bounce preconditioning(See 3.6) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce pre-conditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified requirements(See 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection - Each unit of each lot of equipment shall be inspected for conformance to all the inspection test requirements of Table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government except the overall operation inspection shall be last.

4.5.2 Group B Inspection - This inspection, including sampling, shall conform to Paragraphs 3.8 and 4.12 and to the procedures for small sample inspection of MIL-STD-105, using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have been subjected to and met the Group A inspection.

4.5.3 Group C Inspection - This inspection shall consist of the tests specified in Table III and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with Paragraph 4.5.3.1.

4.5.3.1 Sampling for Inspection of Equipment - Production sampling for Group C inspection shall be two(2) items selected from the first twenty(20) items produced. Three(3) items from each five-hundred(500) produced thereafter shall be subjected to Group C Inspection.

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4.5.3.1(Cont'd)

When preproduction samples are specified by contract or purchase order, Group C inspection will not be required for two(2) items of the first twenty(20) production units.

TABLE II - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.8	4.8	1%	4%
Performance	3.4	4.9		
Amplifier	3.4.1	4.9.1		
Listen Position (Privacy Line)	3.4.1.1	4.9.1.1		
Listen Position	3.4.1.2	4.9.1.2	1%*	
Talk Position	3.4.1.3	4.9.1.3	Combined	
Total Noise	3.4.1.4	4.9.1.4	for the	
Input Power	3.4.1.5	4.9.1.5	group	
Loudspeaker	3.4.2	4.9.2		
Voice Coil Impedance	3.4.2.1	4.9.2.1		
Response	3.4.2.2	4.9.2.2	1%*	
Harmonic Distortion	3.4.2.3	4.9.2.3	Combined	
Microphone Output	3.4.2.4	4.9.2.4	for the	
Dielectric Strength	3.4.2.5	4.9.2.5	group	
Buzzer	3.4.4	4.9.3		
Dielectric Strength		4.9.3.1	1%*	
Insulation Resistance		4.9.3.2	Combined	
Acoustic Output		4.9.3.3	for the group	
Overall Operation	3.4.5	4.9.4	1%	4%

*All electrical defects are considered major.

TABLE III - GROUP C INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph
Temperature	3.5.2	4.10
Moisture-Resistance	3.5.3	4.11

NOTE: The speaker and wood cabinet shall not be subjected to the moisture-resistance test.

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TABLE IIIA - GROUP C INSPECTION TEST SEQUENCE

Unit 1	Unit 2
Temperature	Moisture-Resistance

4.5.3.1.1 The contractor shall perform an Equipment Verification Review(EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction(Para. 3.2). The EVR shall consist of the following:

4.5.3.1.2 An audit to establish that "as-built" equipment(unit, assembly, module, or part) is in accordance with the cited and product drawings.

4.5.3.1.3 An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements(other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.5.3.1.4 The contractor shall compile an EVR report(s) containing the following information:

a. Identification of unit, assembly, module, or part and details proving compliance with 4.5.3.1.2 and 4.5.3.1.3 above.

b. Discrepancies noted.

c. Corrective action taken.

4.5.3.1.5 The contractor shall make available to the Government copies of his EVR report(s) no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the ECR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

a. Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

b. As required, responsible personnel from each functional department available for discussions in their respective areas.

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4.5.3.1.5(Cont'd)

c. Adequate administrative support for the EVR.

4.5.3.2 Noncompliance - The contractor shall immediately report in writing each Group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection(all inspection, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.6 Reinspection of Conforming Group C Sample Units - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.7 Bounce Preconditioning - The unit, with shock mounts, if any, removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 5 minutes. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subject to acceptance inspection.

4.8 Visual and Mechanical Inspection - The equipment shall be examined for the defects listed in MIL-STD-454.

4.9 Performance - All intercommunication stations shall be operated with an input voltage of 115 VAC except the LS-138( )/F1 which shall have a complete test made at both 115 VAC and 230 VAC input. The test equipment shall conform to MIL-C-45662.

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4.9.1 Amplifier - Set the volume control for maximum gain and the audio generator at 1000 Hz, increase the audio generator output until 2 watts is delivered across the load resistor of the amplifier output. Hold the output voltage of the audio generator constant(at the value required to obtain 2 watts at 1000 Hz) and change the frequency from 200 to 8000 Hz. Measure the frequency response and the 1000 Hz gain. With the volume control set at maximum gain and the audio generator adjusted to maintain 2 watts(all frequencies) across the load resistor of the amplifier output, measure the distortion. The frequency response and the distortion shall be measured at each 1000 Hz increment of the required frequency range. The above test shall be performed on 4.9.1.1, 4.9.1.2, and 4.9.1.3. Preset the impedance switch to the 500 ohm position and perform the tests of 4.9.1.2 and 4.9.1.3, then set the switch to the 50 ohm position and repeat the tests of 4.9.1.2 and 4.9.1.3.

4.9.1.1 Listen Position(Privacy Line) - Place the listen-talk switch in the listen position. Connect an audio generator with a 500 ohm resistor in series to the audio terminals of the chassis connector for the privacy line(privacy line plug disconnected). Connect a 50 ohm load across the audio terminals of the chassis connector for the speaker(speaker disconnected). Measure the output at this point to determine compliance with 3.4.1.1.

4.9.1.2 Listen Position - Place the listen-talk switch in listen position and set the impedance change switch to the predetermined impedance position. Connect an audio generator with a resistor of the same value as indicated by the impedance switch in series to the audio terminals of the chassis connector for the station selector switch (station selector switch plug disconnected). Connect a 50 ohm load across the audio terminals of the chassis connector for the speaker (speaker disconnected). Measure the output at this point to determine compliance with 3.4.1.2.

4.9.1.3 Talk Position - Place the listen-talk switch in talk position and set the impedance change switch to the predetermined impedance position. Connect an audio generator with a resistor series to the audio terminals of the chassis connector for the speaker(speaker disconnected). Connect a load resistor of the same value as indicated by the impedance switch across the audio terminals of the chassis connector for the station selector switch(station selector switch plug disconnected). Measure the output at this point to determine compliance with 3.4.1.3.

4.9.1.4 Total Noise - Place the listen-talk switch in the talk position and set the impedance change switch in the 50 ohm position. With the speaker disconnected, connect a 50 ohm resistor across the

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4.9.1.4(Cont'd)

audio terminals of the chassis connector for the speaker. Connect a 50 ohm resistor across the voice(audio) terminals of the chassis connector for the station selector switch, with the station selector switch disconnected. For any position of the volume control, measure the output voltage for compliance with 3.4.1.4.

4.9.1.5 Input Power - The input power shall conform to 3.4.1.5.

4.9.2 Loudspeaker - See NOTE for loudspeaker below Paragraph 3.4.3.

4.9.2.1 Voice Coil Impedance - The loudspeaker voice coil shall be connected in series with a variable resistance to a suitable signal source which delivers approximately 2 volt amperes to the loudspeaker at 1000 Hz. The voltage drops across the loudspeaker voice coil and variable resistance shall be recorded. The resistance shall be varied until the voltage drops are equal. The 1000 Hz voice coil impedance shall then be considered numerically equal to the value of the resistance. It shall comply with 3.4.2.1.

4.9.2.2 Response - The loudspeaker shall be mounted in a rectangular case with internal dimensions not to exceed 7 inches high, 3-1/4 inches in depth and 7 inches in width, ±1/16 inch. The case shall be fabricated of wood, plastic or metal and shall have a circular opening 4-3/4 inch diameter, ±1/16 inch, cut both in the front and rear faces. The loudspeaker shall be placed at a distance of 3 feet from a calibrated microphone in a substantially free sound field. With the voice coil voltage held constant at 9.5 volts the frequency shall be varied over the frequency range of 300 to 6000 Hz and a recording made of the sound pressure. The response shall conform to 3.4.2.2.

4.9.2.3 Harmonic Distortion - With the loudspeaker mounted in the case of 4.9.2.2 and with the voice coil voltage held constant at 9.5 volts, the loudspeaker harmonic distortion shall be measured with a distortion measuring circuit having and inherent harmonic distortion of less than 0.5 percent. It shall comply with 3.4.2.3.

4.9.2.4 Microphone Output - The loudspeaker shall be mounted in the case of Paragraph 4.9.2.2 and the loudspeaker voice coil loaded with a 50 ohm resistive load. When placed in a substantially free sound field of 10 dynes per sq. cm. at frequencies from 300 to 6000 Hz the voltages measured across the load shall conform to 3.4.2.4.

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4.9.2.5 Dielectric - The loudspeaker shall withstand a peak voltage of 100 volts 60 Hz a.c. applied between voice coil and frame for a period of one minute with a simultaneous excitation of the voice coil by a two watt 60 Hz signal (See 3.4.2.5).

4.9.3 Buzzer - The buzzer shall be tested as a separate unit.

4.9.3.1 Dielectric Strength - The buzzer shall withstand an alternating potential of 450 volts r.m.s. applied between the windings and core for 15 seconds. There shall be no evidence of breakdown or degradation.

4.9.3.2 Insulation Resistance - The insulation resistance of the buzzer shall be not less than 50 megohms when measured between the windings and the core with a d.c. potential of not less than 100 volts, applied for not less than 1 minute.

4.9.3.3 Acoustic Output - The buzzer shall be suspended in air from a flexible support and operated in the absence of extraneous noise. The buzzer shall produce a continuous signal clearly audible at a minimum distance of 25 feet.

4.9.4 Overall Operation - The following tests shall be made with the test signal connected to the external terminal board and each individual circuit tested.

a. Set the impedance switch to the 50 ohm position and apply an audio test signal with a 50 ohm resistor in series to a voice terminal and perform the test.

b. Apply an audio test signal with a 500 ohm resistor in series to the privacy line and perform the test.

c. Activate the signal circuit by connecting a 30 ohm resistor across it and perform the test.

d. All the above tests shall conform to 3.4.5.

4.10 Temperature Test for Equipment - The equipment shall be subjected to the temperature cycle shown in MIL-STD-169. Measurements shall be taken as follows after a warm up period of one hour:

Step	Paragraph	Performance
1	4.9.1 & 4.9.2	Full
4	4.9.1 & 4.9.2	Specification
-25°F	4.9.1 & 4.9.2	Requirements
10	4.9.1 & 4.9.2	

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4.10.1 Failure - If the equipment fails to meet the performance established by Step 1 during the other steps where measurements are taken, it shall be realined or readjusted once. If the equipment then fails to meet the performance established in Step 1 or fails subsequently during cycling, it does not pass the test.

4.11 Moisture-Resistance Test for Equipment - The equipment shall be tested as follows:

4.11.1 Test Conditions -

a. Do not move components from the humidity chamber for measurements.

b. Start measurements one hour after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.11.2 Test Method -

a. Dry at 130°F ±5°F for 24 hours.

b. Condition at 77°F ±5°F and 40 to 50 percent relative humidity for 24 hours.

c. Take measurements as specified in 4.11.3, and readjust or realine as necessary to meet full specification requirements.

d. Subject to continuous cycling for two 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.11.3 shall be made during the periods shown in the standard and shall comply with 4.11.3.

e. After cycling has been completed, condition the equipment for 6 hours, at 77°F ±5°F and 40 to 60 percent relative humidity; then adjust for optimum performance, using only those means provided by the components. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specification requirements for those measurements specified in 4.11.3.

4.11.3 Performance - Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	Measurement	Performance During Cycling
4.9.1	Amplifier	Full Specification Requirements
4.9.2	Loudspeaker	Full Specification Requirements

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4.11.4 Failure - If any component fails to meet the performance specified in 4.11.3 during cycling, it shall be realined or readjusted once. If the components then fails to meet 4.11.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements as specified in 4.11.2(e), it does not pass the test.

4.12 Inspection for Dimensional Interchangeability - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating, of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.13 Quality Conformance Inspection of Preparation for Delivery - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance with Section 5.

NOTE: The equipment shall be shipped with the impedance change switch in the 50 ohm position.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging shall be Level A or C as specified (See 6.1c).

5.1.1 Level A -

5.1.1.1 Cleaning - Each intercommunication station shall be cleaned in accordance with Process C-1 of MIL-P-116.

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5.1.1.2 Drying - Each intercommunication station shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation Application - None required.

5.1.1.4 Unit Packaging - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as prescribed herein.

5.1.1.4.1 Technical Literature - Each technical literature shall be Packaged Method IC-1,

5.1.1.4.2 Fuses - All fuses shall be Packaged Method III as follows: Cushion fuses bearing the same stock number together in paperboard conforming to PPP-P-291, Type I. Secure wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A.

5.1.1.4.3 Lamps - Each lamp shall be individually Package Method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, Type I. Secure wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A.

5.1.1.4.4 Electron Tubes - All tubes not securely clamped in place shall be removed and, with the spare tubes, shall be individually packaged in accordance with MIL-E-75.

5.1.1.4.5 Consolidated Package - Consolidate the items, packaged as specified in 5.1.1.4.2 through 4.1.1.4.4, within a close-fitting fiberboard box conforming to PPP-B-636, W6c. Fill all voids with paperboard conforming to PPP-P-291, Type I, to prevent movement. Close the box in accordance with the appendix of the box specification.

5.1.1.4.6 Intercommunication Stations - Each intercommunication station shall be individually Packaged Method IA-14 as follows: Wind each cable assembly into a coil of proportionate dimensions and tie in three equidistant places with cotton twine conforming to T-T-871. Wrap each junction box with paperboard conforming to PPP-P-291, Type III, and secure wrapping with gummed paper tape conforming to PPP-T-45, Type III, Grade A. Secure each cable assembly, junction box and telephone to the station by means of pressure-sensitive tape conforming to PPP-T-76. Cushion each station on all surfaces with cells or pads of both fabricated of fiberboard conforming to PPP-F-320, Type CF, class domestic, variety SW, Grade 200. Place the cushioned item together with the consolidated package, packaged as specified in 5.1.1.4.5, within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, class domestic, variety SW. Close the box with tape conforming to PPP-T-45, Type III. Blunt all the corners of the box. All stapled or

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5.1.1.4.6(Cont'd)

stitched joints shall be covered with pressure-sensitive tape conforming to PPP-T-76. Place the boxed item within a barrier bag fabricated of material conforming to MIL-B-131, Class 1. Head seal the water-vapor proof barrier bag fabricated of material conforming to MIL-B-131, Class 1. Head seal the water-vapor proof barrier bag, leaving a small opening to exhaust the air. Exhaust the entrapped air and complete the seal. Place the bagged item within a close-fitting fiberboard box conforming to PPP-B-363, W5c. Any stapling or stitching on the inside of the box shall be covered with pressure-sensitive tape conforming to PPP-T-76. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the bagged item directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C - Intercommunication stations shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing - Packing shall be Level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without amaging the container or its contents.(See 6.1c).

5.2.1 Level A

5.2.1.1 Palletized Load - A quantity of intercommunication stations bearing the same stock number and packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less Than Palletized Load - When quantities per destination are less than a pallet load, the items packaged as specified in 5.1 and bearing the same stock umber, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproffed contai ers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids laid flat shall be applied in

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5.2.1.2(Cont'd)

accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type I, Class B.

5.2.2 Level B - A quantity of intercommunication stations, bearing the same stock number and packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.1 Less Than Palletized Load - When quantities per destination are less than a pallet load, a quantity of intercommunication stations, bearing the same stock number and packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Style E, or PPP-B-636, Type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to MIL-P-15011 or MIL-P-3938. Closure shall be in accordance with the appexdix of the applicable box specification. Reinforcing shall be by nonmentallis strapping or pressure-sensitive failament tape banding conforming to PPP-S-760, Type II and PPP-T-97, Type IV, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C -

5.2.3.1 Palletized Load - A quantity of intercommunication stations, bearing the same stock number and packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less Than Palletized Load - When quantities per destination are less than a pallet load, a quantity of intercommunication stations, bea ing the same stock number and packaged as specified in 5.1, shall be packed as specified in 5.2.2.1 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, Class I and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering Data - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. Level of preservations and packing and packaging(See Section 5).

d. When the rough handling and functional tests are required (See Section 5).

e. The specific paragraphs of Section 5 which are applicable to the particular procurement.

f. Preproduction inspection:

Five sample units of the required item cited in Section 1 are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on three sample units(See 3.1).

g. Marking and shipping of samples.

h. Place of final inspection.

i. Submission of the statement of treatment referenced in 3.3.1, as soon as possible after award of contract. This statement should be submitted to the Contracting Officer.

6.2 Nomenclature - The parentheses in the nomenclature will be deleted or replaced by a letter indentifying the particular design; for example LS-125X/Fl. The contractor should apply for nomenclature in accordance with the applicable clause in the contract(See 1.1).

6.3 Group C Inspection - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the Contracting Officer on the adequacy of corrective action (See 4.5.3.2).

6.4 Verification Inspection - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality

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6.4(Cont'd)

assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Project Number
5830-A026

Custodian
Army-EL

Preparing Activity
Army-EL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-I-12637B(EL), Intercommunication Stations LS-124( )/F1, LS-125( )/F1, -126, -127, -128, -138, -200, -201, -202

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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